

# Dingyang Chen

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## Education

- PhD**, Computer Science May 2021 - May 2025  
**University of South Carolina**, Columbia, SC, USA  
Advisor: Qi Zhang
- MEng**, Electrical Engineering and Computer Sciences Aug 2018 - May 2019  
**University of California, Berkeley**, Berkeley, CA, USA
- BS**, Computer Science Aug 2014 - May 2018  
**University of Illinois at Urbana-Champaign**, Champaign, IL, USA

## Professional Experience

- Amazon**, Seattle, WA Jan 2025 - Present  
Applied Scientist
- Fine-tuned large language models to automate product quality detection at Amazon scale.
- Amazon**, Bellevue, WA May 2024 - Aug 2024  
Applied Scientist Intern
- Developed multi-armed bandit models for cost-of-goods prediction.
- Siemens**, Remote May 2022 - Aug 2022  
Research Intern
- Applied reinforcement learning algorithms to optimize power system reconfiguration.

## Publications

### Journal Articles

- [1] **Analyzing Best-Response Dynamics for Cooperation in Markov Potential Games**  
Dingyang Chen, Xiaoling Zeng, Thinh T. Doan, Qi Zhang  
Transactions on Machine Learning Research (TMLR), 2026.

### Refereed Conference Papers

- [1] **Natural Policy Gradient for Bayesian Network Policies in Markov Potential Games.**  
Dingyang Chen, Zhenyu Zhang, Yuan Ling, Qi Zhang  
Reinforcement Learning Conference (RLC), 2026.
- [2] **Correlated Policy Optimization in Multi-Agent Subteams.**  
Dingyang Chen, Jianing Ye, Zhenyu Zhang, Xiaolong Kuang, Xinyang Shen, Ozalp Ozer, Chongjie Zhang, Qi Zhang  
International Conference on Learning Representations (ICLR), 2026.
- [3] **Reinforcement Learning with Euclidean Data Augmentation for State-Based Continuous Control.**  
Jinzhu Luo, Dingyang Chen, Qi Zhang  
Conference on Neural Information Processing Systems (NeurIPS), 2024.

- [4] **Efficient Sequential Decision Making with Large Language Models.**  
Dingyang Chen, Qi Zhang, Yinglun Zhu  
 Empirical Methods in Natural Language Processing (EMNLP), 2024.
- [5] **E(3)-Equivariant Actor-Critic Methods for Cooperative Multi-Agent Reinforcement Learning.**  
 Dingyang Chen, Qi Zhang  
 International Conference on Machine Learning (ICML), 2024.
- [6] **Context-Aware Bayesian Network Actor-Critic Methods for Cooperative Multi-Agent Reinforcement Learning.**  
Dingyang Chen, Qi Zhang  
 International Conference on Machine Learning (ICML), 2023.
- [7] **Communication-Efficient Actor-Critic Methods for Homogeneous Markov Games.**  
 Dingyang Chen, Qi Zhang  
 International Conference on Learning Representations (ICLR), 2022.

### **Refereed Workshop Papers**

- [1] **From Business Metrics to Behavioral Personas Controllable User Simulation for Pre-Deployment Agent Testing.**  
 Zhenyu Zhang, Yuan Ling, Dingyang Chen, Xinyang Shen  
 Workshop on Agents in the Wild at ICML, 2026.
- [2] **Tree of Options: Temporally Extended World Modeling, Planning, and Execution with Large Language Models.**  
 Xiaoling Zeng, Dingyang Chen, Qi Zhang  
 Workshop on World Models at ICLR, 2026.
- [3] **Convergence Rates of Bayesian Network Policy Gradient for Cooperative Multi-Agent Reinforcement Learning.**  
Dingyang Chen, Zhenyu Zhang, Xiaolong Kuang, Xinyang Shen, Ozalp Ozer, Qi Zhang  
 Workshop on Bayesian Decision-making and Uncertainty at NeurIPS, 2024.
- [4] **Convergence and Price of Anarchy Guarantees of the Softmax Policy Gradient in Markov Potential Games.**  
Dingyang Chen, Qi Zhang, Thanh T. Doan  
 Decision Awareness in Reinforcement Learning Workshop at ICML, 2022.
- [5] **A Meta-Gradient Approach to Learning Cooperative Multi-Agent Communication Topology.**  
 Qi Zhang, Dingyang Chen  
 Workshop on Meta-Learning at NeurIPS, 2021.

### **Professional Service**

#### **Conference Reviewer**

ICML 2024 & 2025 & 2026, AISTATS 2026, AAAI 2025, ICLR 2024, NeurIPS 2023, 2025, and the NeurIPS 2022 Deep RL Workshop

#### **Journal Reviewer**

IEEE TNNLS

### **Teaching Experience**

**Graduate Teaching Assistant** - Visualization Tools, University of South Carolina [FA24]  
**Graduate Teaching Assistant** - Algorithmic Design I, University of South Carolina [FA23]